Quantitative Methods 1

Chapter 2: Measures of central tendency

Exercise worksheet

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Exercise 4 — Centre and unusual observations

Seven customer baskets (euros): $16, 18, 18, 20, 22, 25, 91$.

1. Calculate the mode, median and mean.
2. Recalculate the mean excluding the last observation and compare.
3. Choose the most appropriate indicator of a typical basket.

Exercise 5 — Selecting the appropriate average

A company has grades $12, 15, 17$ weighted $20\%, 30\%, 50\%$. Its revenue grows by $+8\%, -5\%, +10\%$ over three years. A courier travels 80 km at 40 km/h, then 80 km at 80 km/h.

1. Calculate the weighted average grade.
2. Calculate the compound average annual growth rate.
3. Calculate the average speed over the 160 km.
4. Explain why the ordinary arithmetic mean is not appropriate in every situation.

Exercise 6 — Typical price and a pricing decision

Use the 150-consumer portable solar charger distribution from Exercise 3. The full table is reproduced below so that this worksheet can be used independently.

Class (euros)	12–18	18–24	24–30	30–36	36–45	45–60	60–75
Count n_i	6	14	28	42	30	20	10

1. Find the modal class using densities.
2. Find the median and quartiles by linear interpolation.
3. Estimate the mean using class midpoints.
4. Find the highest price affordable to at least 60

General instructions: justify formulas, display intermediate calculations, and interpret results in their business context. Grouped-data calculations yield estimates.